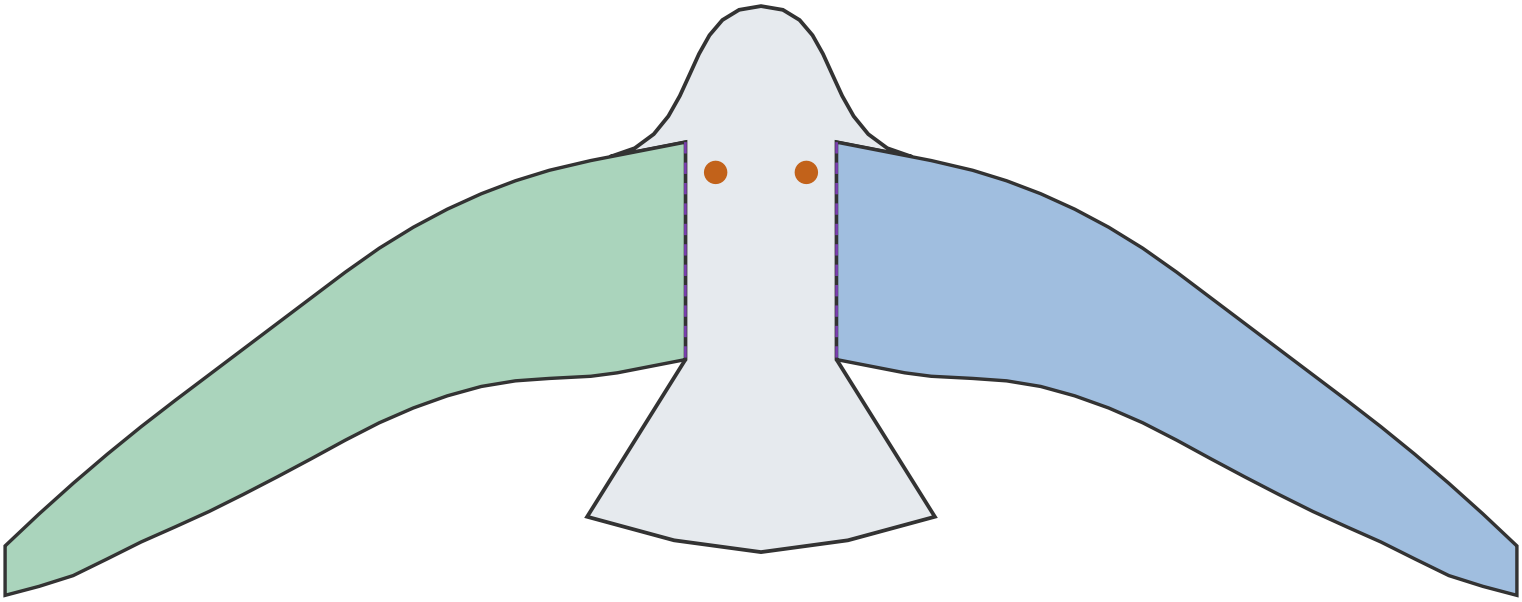
Folding concept 1: geared single-pivot swing wing

Swift outline A, 400 mm foam build, top view, nose up. Printed at 100 % this sheet is half size.

Concept 1: open (glide pose)

joint angles 0 deg

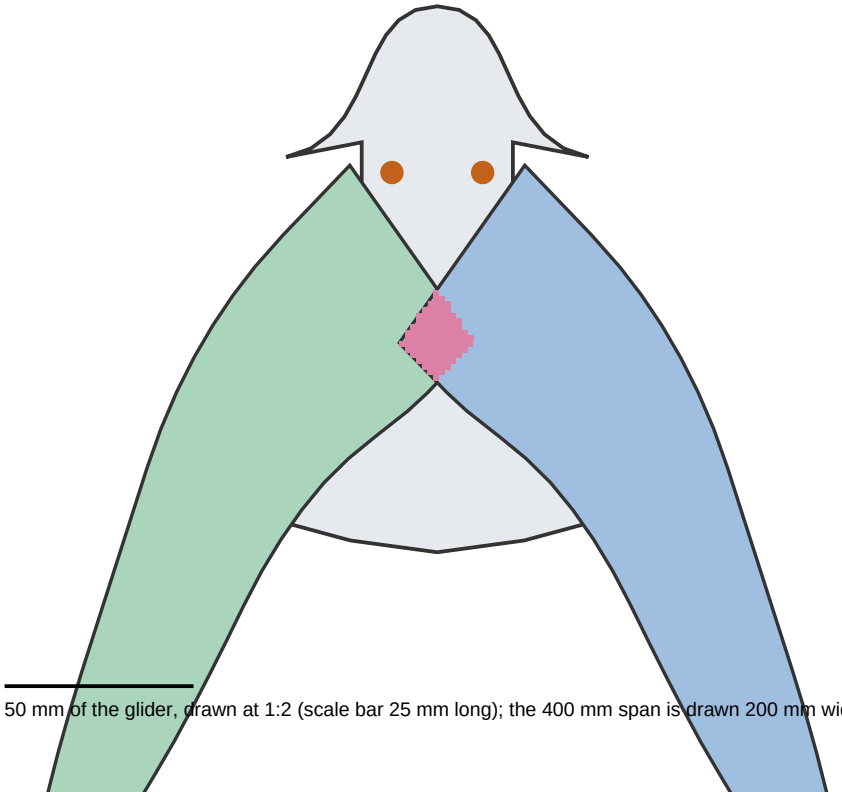


span 400 mm, length nose to tail end 144 mm; dashed purple = fold lines; amber dashed = path of the tip

orange dots = pivot pins; green dot (folded view) = hook

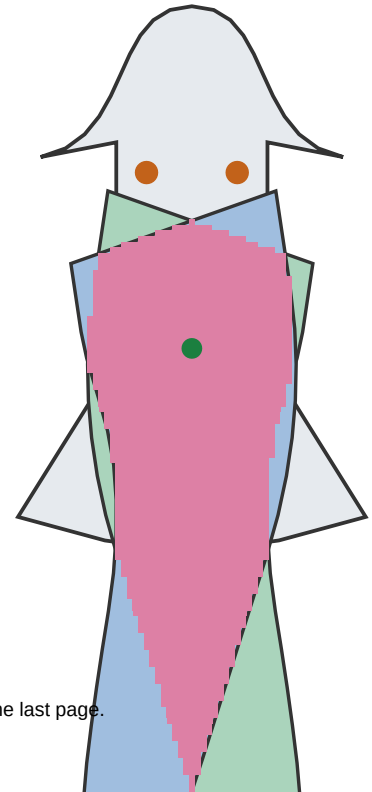
Concept 1: part-folded (half way)

shoulder 35 deg



Concept 1: fully folded (stowed)

shoulder 70 deg



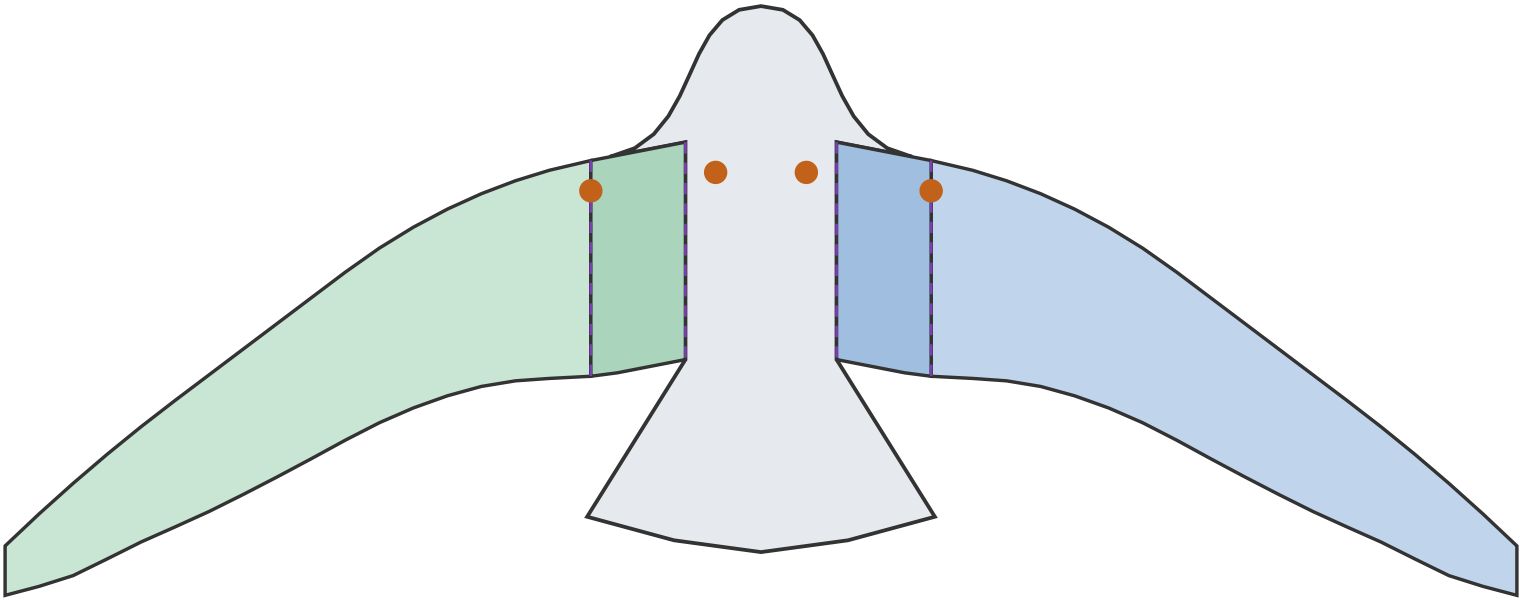
50 mm of the glider, drawn at 1:2 (scale bar 25 mm long); the 400 mm span is drawn 200 mm wide. Stowed states at 1:1 on the last page.

Folding concept 2: two pieces, reversed string

Swift outline A, 400 mm foam build, top view, nose up. Printed at 100 % this sheet is half size.

Concept 2: open (glide pose)

joint angles 0 deg

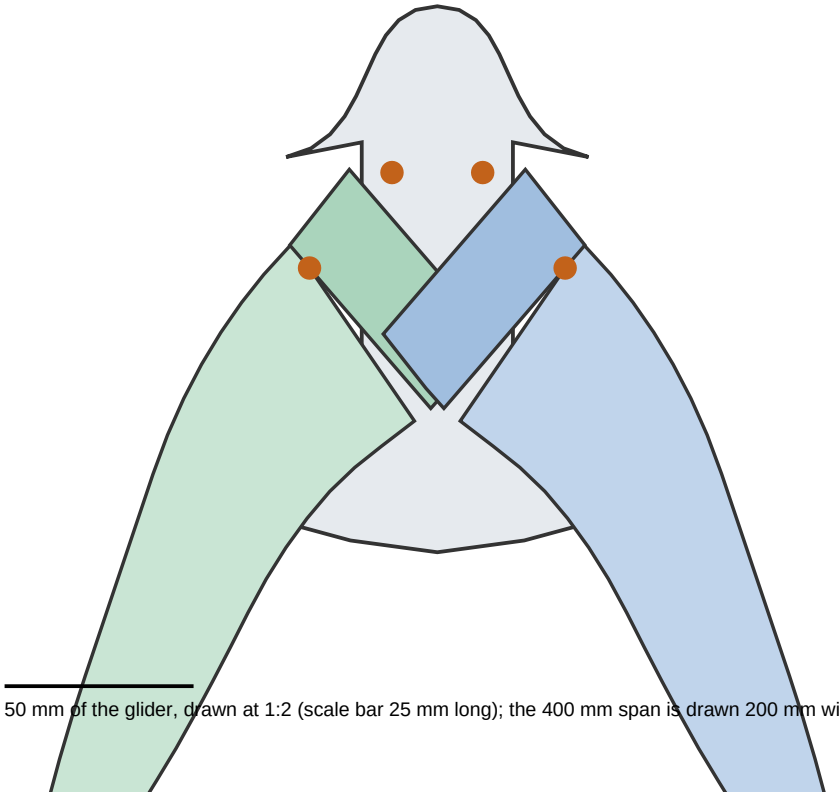


span 400 mm, length nose to tail end 144 mm; dashed purple = fold lines; amber dashed = path of the tip

orange dots = pivot pins; green dot (folded view) = hook

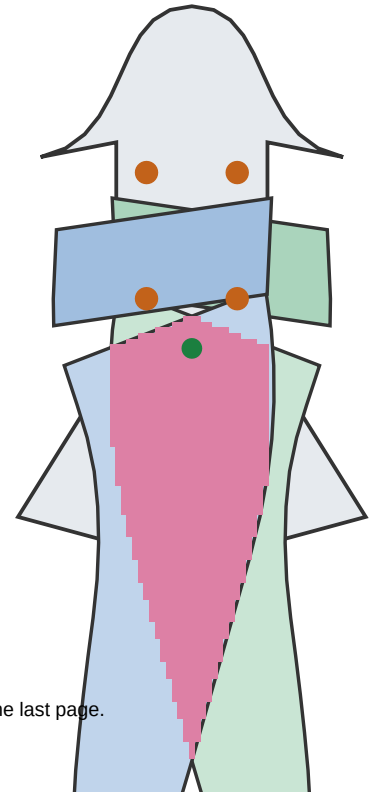
Concept 2: part-folded (half way)

shoulder 41 deg wrist -6 deg



Concept 2: fully folded (stowed)

shoulder 82 deg wrist -13 deg

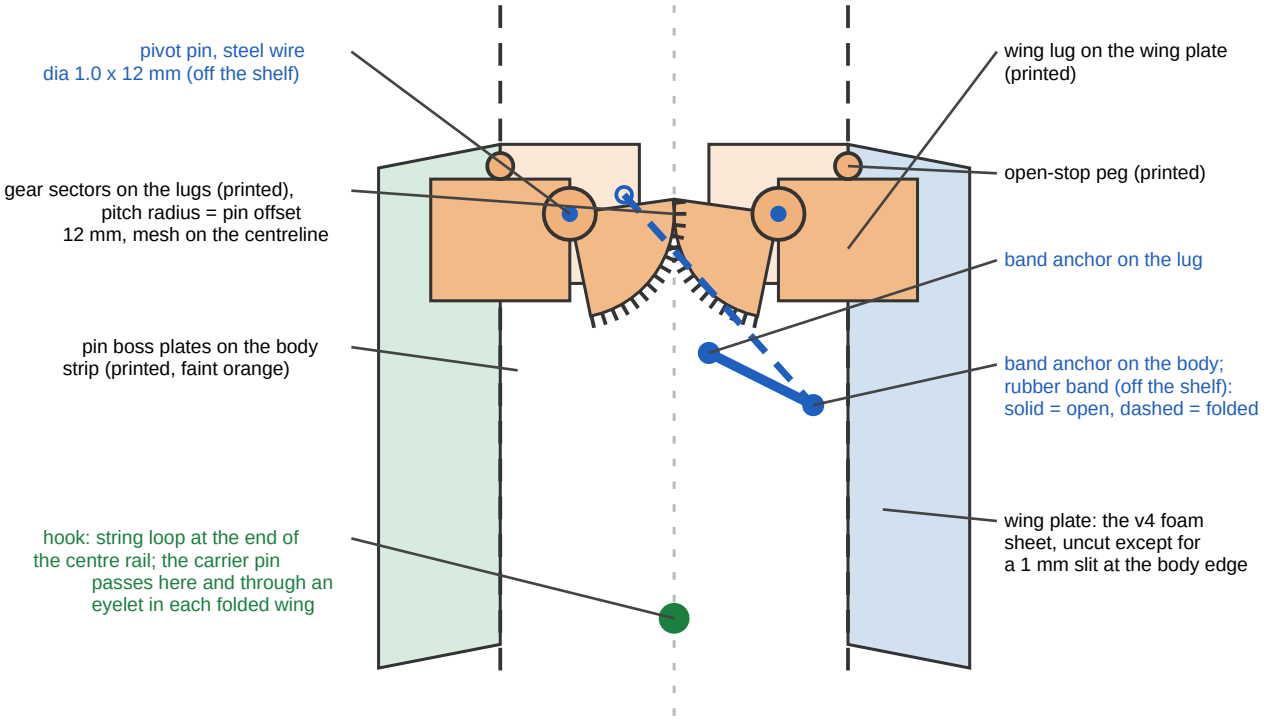


50 mm of the glider, drawn at 1:2 (scale bar 25 mm long); the 400 mm span is drawn 200 mm wide. Stowed states at 1:1 on the last page.

Mechanism sketch, concept 1

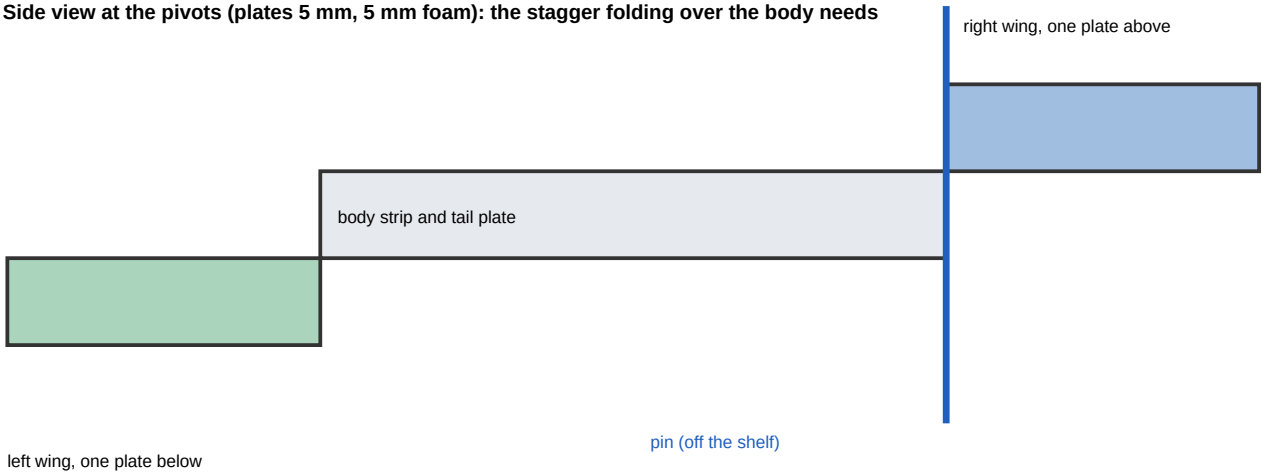
Scale about 1.15:1 (not 1:1). Orange = printed, blue = off the shelf, green = hook.

Concept 1: geared swing wing, band-opened: root of the glider, open pose, nose up



orange = printed blue = off the shelf (pin, band) green = hook / hold point

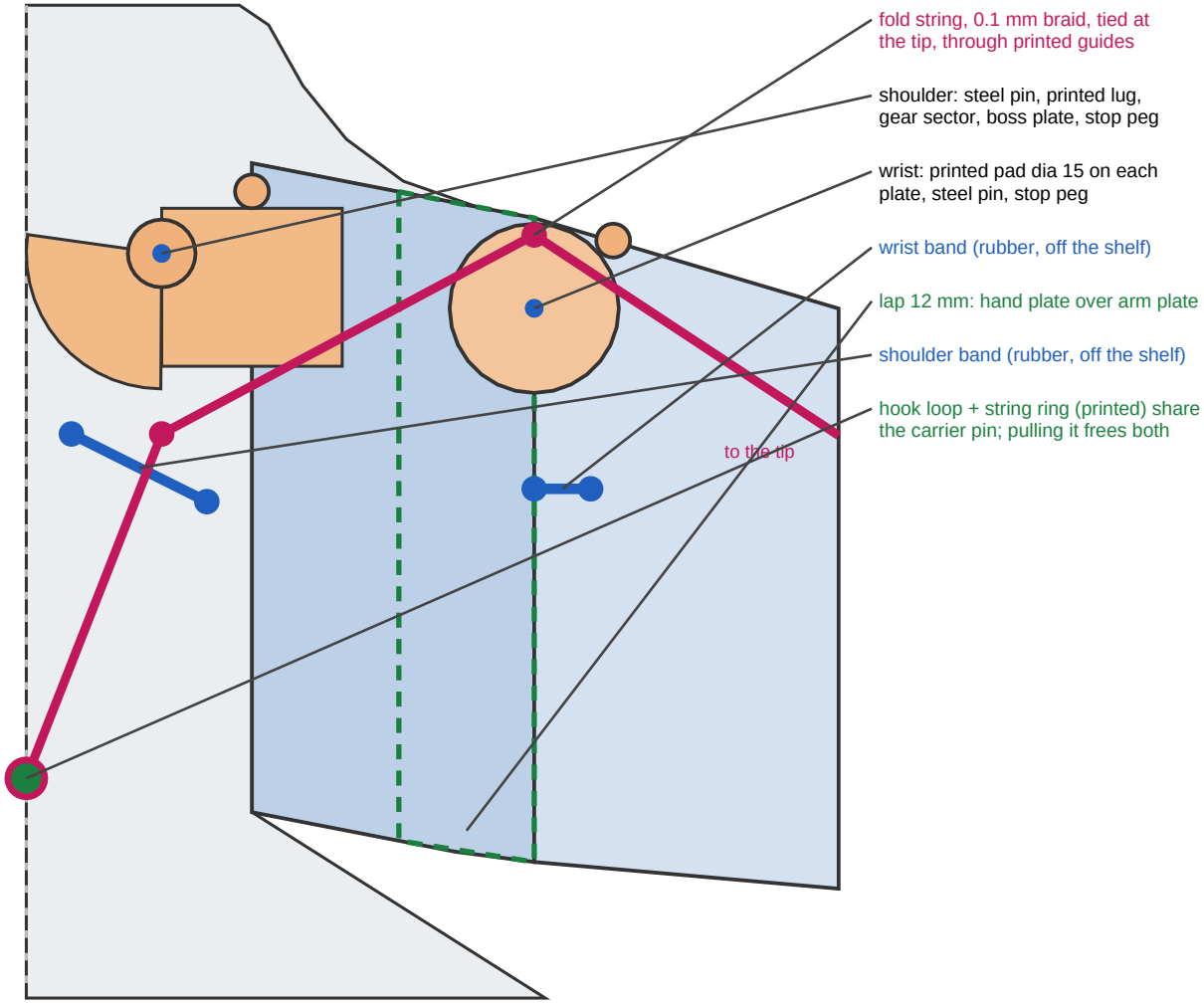
Side view at the pivots (plates 5 mm, 5 mm foam): the stagger folding over the body needs



Mechanism sketch, concept 2

Scale about 1.5:1 (not 1:1). Orange = printed, blue = off the shelf, magenta = fold string, green = hook.

Concept 2: reversed string, two pieces per side: root and wrist of the right half, open pose, nose up

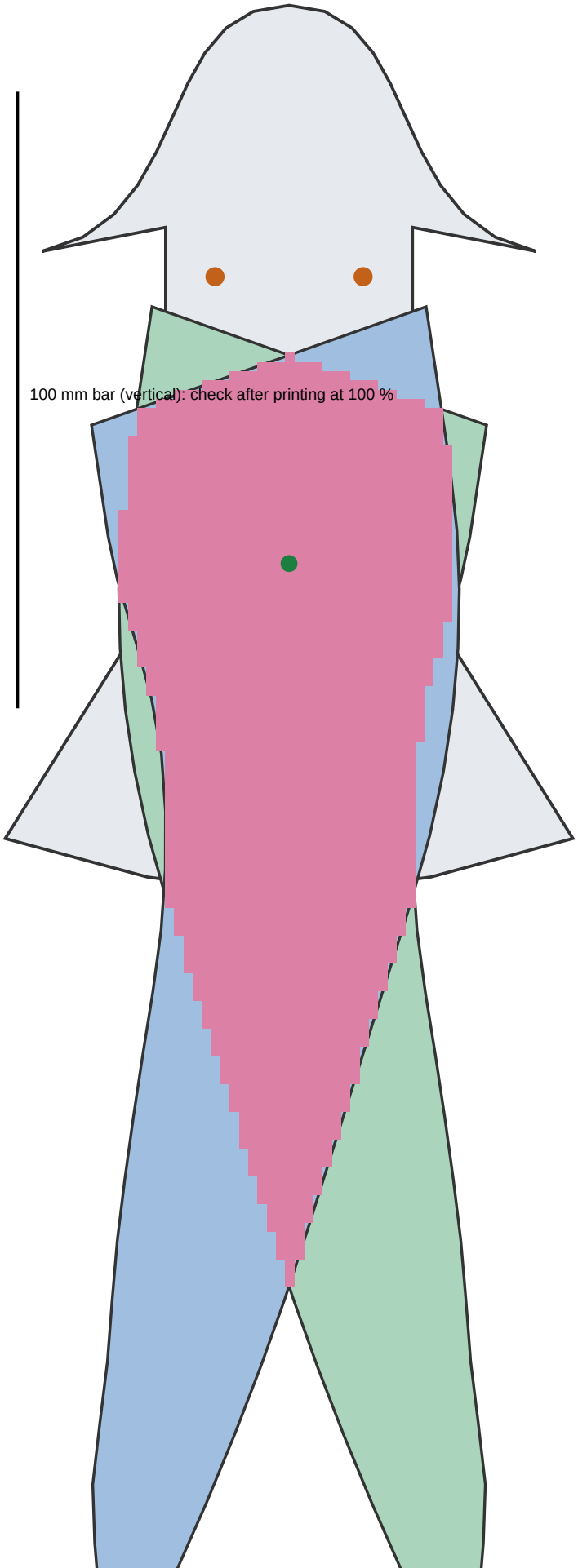


orange = printed blue = off the shelf magenta = fold string green = hook (band anchor positions: schematic)

Stowed states at 1:1

Print at 100 %. Green dot = hook; orange dots = pins; pink = left / right wing overlap.

Concept 1 folded, 1:1



Concept 2 folded, 1:1

